

**UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
DHAKA, BANGLADESH



PROJECT PROPOSAL

**The Round Table Exchange (RTE): A Multi-Party Skill Barter Network
with Graph-Based Cycle Matching**

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Table of Contents

- 1. Objectives 1
 - 1.1 Primary Objective..... 1
 - 1.2 Secondary Objectives..... 1
 - 1.3 Explicit Non-Objectives..... 1
- 2. Scope 2
 - 2.1 Project Context & Platform Paradigms 2
 - 2.2 In-Scope Components 2
 - 2.3 Out-of-Scope (Future Work) 2
- 3. Feasibility..... 3
 - 3.1 Technical Feasibility..... 3
 - 3.2 Time Feasibility 3
 - 3.3 Resource Feasibility 3
- 4. WBS (Work Breakdown Structure) 4
 - 4.1 Detailed Work Breakdown Structure..... 4
 - 4.2 Team Role Allocation 5
- 5. Gantt Chart (Project Timeline) 7
 - 5.1 Project Timeline (12 Weeks)..... 7
- 6. Budget 8
 - 6.1 Project Budget (Bangladeshi Market Context) 8
 - 6.1.1 Cloud Infrastructure & Hosting Costs..... 8
 - 6.1.2 Development Support & Data Benchmarking 8
 - 6.1.3 Total Budget Allocation Summary 8

1. Objectives

1.1 Primary Objective

The main goal of this project is to **build a graph-based multi-party matching engine for skill bartering**. When users post skills they can teach and skills they want to learn, the system represents these requests as a graph and automatically finds exchange loops among 3 or more people.

1.2 Secondary Objectives

1. **Simple Web Application:** Build a simple web app where users can create profiles, offer skills, and request skills with location and time availability.
2. **Trade Confirmation System:** Build an accept/reject workflow where all matched users in a trade loop can accept or reject a trade proposal.
3. **Synthetic Test Data Generator:** Build a fake data generator to test how well the matching engine works as the number of users grows.

1.3 Explicit Non-Objectives

To keep the project doable within one semester, the following features are not part of this project:

- User trust and reputation rating algorithms.
- Legal or trade dispute resolution tools.
- SMS/Push notifications and calendar scheduling.
- Advanced graph databases like Neo4j.

2. Scope

2.1 Project Context & Platform Types

Existing skill exchange platforms usually work in one of two ways:

1. **Direct 1-to-1 Trade Apps (e.g., Reciproc8, Ying, BarterQuest, The Barter Shop):** Require two people to want exactly what the other has at the same time.
2. **Time Bank Apps (e.g., TimeRepublik, hOurworld):** Use time credits as virtual money, so matching is not needed.

The Round Table Exchange (RTE) fills the gap by finding multi-person trade loops automatically, inspired by kidney donation matching algorithms [1, 9] and multi-party trade research [2, 7].

2.2 In-Scope Features

- **Matching Engine:** Directed graph search algorithm that finds trade loops up to 4 people and ranks them by location and schedule overlap.
- **Basic Web App:** Simple user login, profiles, and skill offer/request listings with location coordinates.
- **Trade Confirmation Workflow:** Multi-user accept/reject system with proposal time limits.
- **Test Data Generator:** Configurable tool to generate fake user data for testing.

2.3 Out-of-Scope Features

- User reputation and review scores.
- Dispute resolution mechanisms.
- Real-time push notifications and calendar sync.
- Dedicated graph database software (e.g., Neo4j).

3. Feasibility

3.1 Technical Feasibility

Finding cycles in directed graphs is a well-studied problem. For small loop sizes (up to 4 people), a Depth-First Search (DFS) algorithm works fast and efficiently for up to 1,000 users [8]. Using a synthetic data generator ensures we can test and demonstrate the system easily even without real users.

Table 3.1: Summary of Technical Feasibility

Risk Area	Solution	Feasibility Level
Algorithm Speed	Bounded DFS algorithm (up to 4 users per loop)	High / Fast
No Real Users Yet	Synthetic data generator creates test users	Fully Solved
Demonstration Risk	Pre-made test data guarantees trade loops for demo	Fully Solved

3.2 Time Feasibility

The project is planned over 12 weeks. Core engine development (Weeks 3–6) happens alongside web app development (Weeks 5–8), leaving final weeks for testing and report writing.

3.3 Resource Feasibility

All software tools used in this project are free and open-source.

Table 3.2: Resource Stack & Tools

Component	Technology Used	Cost
Database	PostgreSQL	Free / Open-Source
Backend	Node.js / FastAPI	Free / Open-Source
Frontend	React.js	Free / Open-Source
Testing & Graphs	Python (NetworkX, NumPy)	Free / Open-Source

4. WBS (Work Breakdown Structure)

4.1 Detailed Work Breakdown Structure

The Round Table Exchange (RTE) project is structured into seven primary Work Packages (WP), further broken down into detailed sub-tasks and implementation deliverables:

1. WP 1.0: Project Management & Governance

- (a) *1.1 Proposal Formulation:* Requirements gathering, scope definition, and formal proposal drafting.
- (b) *1.2 Milestone & Sprint Tracking:* Bi-weekly progress reviews, task allocation, and risk management.
- (c) *1.3 Documentation & Final Reporting:* Technical report compilation, revision control, and presentation prep.

2. WP 2.0: Requirements & Architectural Design

- (a) *2.1 Stakeholder & Use-Case Analysis:* Defining user personas, offerer/requester flows, and admin management.
- (b) *2.2 Relational Data Modeling:* Schema design for Users, Profiles, Skill Tags, Offers, Wants, and Adjacency graphs in PostgreSQL.
- (c) *2.3 Module Interface Specification:* Defining REST/JSON contracts separating the matching engine from backend services.

3. WP 3.0: Core Matching Engine Development (Core Contribution)

- (a) *3.1 Graph Construction Service:* Converting active database listings into directed adjacency graphs.
- (b) *3.2 Bounded-Depth DFS Cycle Finder:* Implementing path search algorithms for closed loops up to length $k = 4$.
- (c) *3.3 Multi-Signal Ranking Engine:* Scoring cycles using Haversine geographic proximity and time-window overlap.
- (d) *3.4 Synthetic Data Generator:* Building a configurable dataset generator (tunable node count, edge density, skill distribution).
- (e) *3.5 Graph Engine Unit Testing Suite:* Unit test cases covering zero-cycle, single 3-cycle, and complex overlapping cycle graphs.

4. WP 4.0: Application Layer & API Services

- (a) *4.1 Authentication & Profile CRUD:* User registration, JWT-based auth, and profile management.
- (b) *4.2 Listing Management Module:* Creating, editing, and disabling skill offer/want posts with tags and location coords.

- (c) *4.3 RESTful API Gateway*: Middleware endpoints connecting web client requests to the graph matching service.

5. WP 5.0: Group Confirmation Workflow Engine

- (a) *5.1 Proposal Dispatcher*: Pushing trade proposal alerts to all matched cycle participants.
- (b) *5.2 Accept/Reject State Machine*: State machine tracking individual responses (Pending, Accepted, Rejected, Expired).
- (c) *5.3 Timeout & Expiration Handler*: Automated cron/background worker canceling proposals if unconfirmed within window.
- (d) *5.4 Re-trigger Matcher Service (Stretch Goal)*: Re-evaluating graph matches upon partial rejection.

6. WP 6.0: System Benchmarking & Evaluation

- (a) *6.1 Scalability Experiments*: Measuring match rate vs. user density across graph sizes (50 to 1,000 nodes).
- (b) *6.2 Runtime Complexity Analysis*: Empirical validation of $O(n \cdot m^{k-1})$ time complexity bounds.

7. WP 7.0: Documentation & Deployment

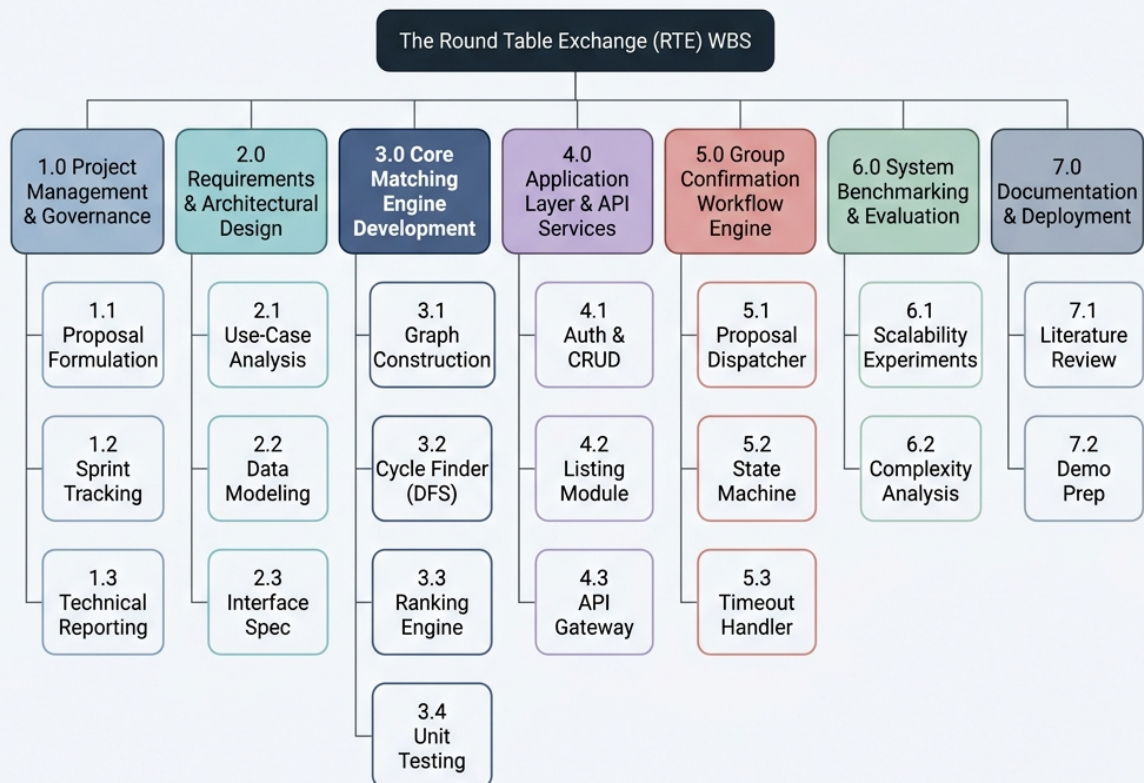
- (a) *7.1 Literature & Mechanism Review*: Finalizing prior-art comparison writeup.
- (b) *7.2 System Integration & Demo Prep*: End-to-end testing, demo script creation, and final report submission.

4.2 Team Role Allocation

Work package ownership and technical responsibilities are structured as follows:

- **Ahmmad Abdali Khan (ID: 0432320005101118)**: Lead Software Engineer. Manages core graph matching algorithm engineering, bounded-depth DFS cycle discovery engine, multi-signal ranking function, PostgreSQL database schema design, RESTful backend API gateway, and empirical runtime complexity analysis (WBS 3.1, 3.2, 3.3, 4.3, and 6.2).
- **Sumaia Bintey Ismail (ID: 0432320005101103)**: Application & Systems Engineer. Manages frontend web application UI/UX, user profile CRUD workflows, skill listing management module, synthetic data generator module, group confirmation workflow state machine, use-case specifications, and final technical documentation (WBS 1.0, 2.0, 3.4, 3.5, 4.1, 4.2, 5.0, 6.1, and 7.0).

The Round Table Exchange (RTE) — Work Breakdown Structure



The Round Table Exchange (RTE) Project Proposal — WBS Diagram

Figure 4.1: Professional Work Breakdown Structure (WBS) Infographic Diagram of Cycle-Trade

5. Gantt Chart (Project Timeline)

5.1 Project Timeline (12 Weeks)

The project timeline spans 12 weeks across the academic semester. The work packages are consolidated into 8 key high-level task groups:

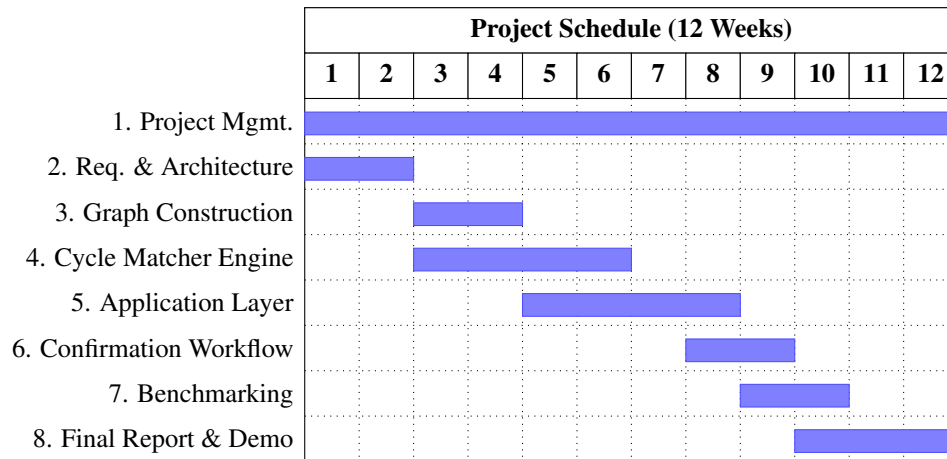


Table 5.1: Task Schedule Summary (8 Tasks across 12 Weeks)

Task #	Task Description	Start Week	End Week
1	Project Management & Documentation	Week 1	Week 12
2	Requirements Analysis & Architecture	Week 1	Week 2
3	Data Schema & Graph Construction Module	Week 3	Week 4
4	Cycle Detection Algorithm & Ranking Engine	Week 3	Week 6
5	Application Layer & REST API Development	Week 5	Week 8
6	Confirmation Workflow State Machine	Week 8	Week 9
7	System Benchmarking & Complexity Analysis	Week 9	Week 10
8	Final Report Writing & Demonstration Prep	Week 10	Week 12

6. Budget

6.1 Project Budget (Bangladeshi Market Context)

The budget for The Round Table Exchange (RTE) is planned for the Bangladeshi market, allocating a total estimated cost of **BDT 50,000** (Fifty Thousand BDT) across hosting, server tools, and testing setup.

6.1.1 Cloud Infrastructure & Hosting Costs

Hosting the web application and matching server for 6 months requires cloud server infrastructure as shown in Table 6.1.

Table 6.1: Cloud Infrastructure & Hosting Costs (6 Months)

Resource / Service	Provider	Total Cost (BDT)
Application Server (VPS for Matching Engine)	DigitalOcean / Cloud VPS	BDT 12,000
Managed Database (PostgreSQL Cloud)	Supabase / Cloud DB	BDT 9,000
Domain (.com / .bd) & Security SSL	Domain Registrar	BDT 2,000
Subtotal Infrastructure Costs		BDT 23,000

6.1.2 Development Support & Data Tools

Development support covers local workstation setup, testing internet connectivity, and synthetic data testing tools.

Table 6.2: Development Support & Data Tools Allocation

Category	Total Cost (BDT)
Development Workstation Setup & Connectivity Support	BDT 15,000
Synthetic Test Data Generation & Benchmarking Tools	BDT 9,500
Subtotal Development Support Costs	BDT 24,500

6.1.3 Total Budget Allocation Summary

The complete project budget distribution is summarized in Table 6.3.

Note: Student development labor is provided as part of academic project requirements. Initial development testing uses free local server tools.

Table 6.3: Master Project Budget Summary (Total: BDT 50,000)

Category	Amount (BDT)	Percentage (%)
Cloud Infrastructure & Hosting (6 Months)	BDT 23,000	46.0%
Development Support & Data Tools	BDT 24,500	49.0%
Contingency Buffer (5%)	BDT 2,500	5.0%
Total Project Budget	BDT 50,000	100.0%

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